Spin-Mediated Acceleration as a Phenomenological Fitting Function for Galaxy Rotation Curves: Updated Analysis of 175 SPARC Galaxies

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Abstract

An angular-momentum-dependent extra-acceleration term of the form

$$a_{\text{extra}}(r) = k \cdot J_{\text{vis}}/(r + r_0)^2$$

is added to the visible-matter contribution to reproduce observed rotation curves of 175 late-type disk galaxies from the SPARC database. The angular momentum J_{vis} is computed from photometric baryonic components only, avoiding the circular reasoning present in an earlier version of this work. The softening radius is set per galaxy as $r_0 = 2 \times r_{\text{half}}$, a choice validated by a systematic scan of 20 parameterisation strategies. Using continuous L-BFGS-B optimisation with a weighted reduced χ^2 objective, the model achieves a mean fractional error of 18.52% (median 15.76%, $\sigma = 14.96\%$), compared with 30.26% in the original version.

A global power-law relation

$$k \propto J_{\text{vis}}^{-0.652}$$

is statistically preferred over a constant-k model ($\Delta\text{AIC} = -141{,}371$), but explains only modest variance ($R^2 \approx 0.16$) and should not be interpreted as a universal law. Approximately 36 out of 175 galaxies (21%) converge to the optimiser lower bound $k \approx 0$, indicating that the baryonic model alone is sufficient for these systems. A direct comparison with the MOND Radial Acceleration Relation (RAR), applied to the same 175 galaxies under the same error metric, yields MOND mean MAE% = 18.24% (median 10.31%); the two approaches achieve similar mean performance while differing in their error distributions.

The fitted coupling constants exceed the Lense-Thirring frame-dragging prediction by 10–14 orders of magnitude. No first-principles derivation from general relativity is claimed. The model does not generalise to galaxy clusters (mean error 72.5%) and is limited to rotationally supported disk galaxies under the axisymmetric assumption. This work is presented as an exploratory phenomenological study; it does not constitute evidence for a new gravitational mechanism or a replacement for dark matter.

Keywords: galaxy rotation curves; phenomenological model; SPARC survey; angular momentum; fitting function; disk galaxies; MOND comparison

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1. Introduction

1.1 The Galaxy Rotation Curve Problem

The observed flatness of galaxy rotation curves is one of the most persistent challenges in extragalactic astrophysics. Under Newtonian gravity, the circular velocity contributed by visible baryonic matter is expected to decline as

$v \propto r^{-1/2}$

beyond the optical disk, following the Keplerian fall-off of an enclosed mass distribution. Instead, observed velocities remain roughly constant — or even continue to rise — well beyond the stellar disk, across galaxy types spanning four decades in luminosity and a wide range of surface brightness and morphology (Rubin, Ford & Thonnard, 1980; van Albada et al., 1985).

1.2 Standard Explanations and Alternatives

The dominant paradigm attributes the excess rotation to cold dark matter (CDM) halos, typically modelled with NFW or Einasto density profiles (Navarro, Frenk & White, 1997). CDM halos reproduce the observed rotation curves when the halo parameters are tuned per galaxy, but the required halo shapes and concentrations are not always consistent with cosmological simulations, and the tight empirical correlations between baryonic and total mass remain difficult to explain naturally within the CDM framework (McGaugh, Lelli & Schombert, 2016).

The leading baryonic alternative is Modified Newtonian Dynamics (MOND; Milgrom, 1983), which introduces a critical acceleration scale $a_0 \approx 1.2 \times 10^{-10}$ m s^{-2} below which the effective gravitational acceleration deviates from the Newtonian inverse-square law. MOND reproduces many individual rotation curves with a small number of free parameters per galaxy and naturally predicts the Radial Acceleration Relation (RAR; McGaugh et al., 2016; Li et al., 2018). The present work does not claim to replace or supersede either CDM or MOND.

1.3 Purpose and Scope of This Study

In an earlier version of this work (Kherki 2025, hereafter V1), a simple extra-acceleration term proportional to visible-matter angular momentum was explored as a phenomenological fitting function for SPARC rotation curves:

$$a_{\text{extra}}(r) = k \cdot \frac{J_{\text{vis}}}{(r + r_0)^2}$$

The functional form was partly inspired by the Lense-Thirring (LT) frame-dragging precession rate $\Omega_{\text{LT}} \propto J/r^3$, which induces a tangential velocity contribution $v_{\text{LT}} = \Omega_{\text{LT}} \times r \propto J/r^2$. However, as demonstrated in Section 5 of the present paper, the fitted coupling constants exceed the bare Lense-Thirring prediction by 10–14 orders of magnitude, and no rigorous derivation of this gap exists. The model must therefore be understood purely as an empirical fitting function. This conclusion was already noted in V1 but is made more explicit and quantitative here.

The present version (V3) corrects three methodological problems identified in V1: (i) the angular momentum was computed from the observed velocity curve, introducing circular reasoning; (ii) the softening radius r_0 was fixed at 0.5 kpc for all galaxies, an arbitrary choice that is not optimal; and (iii) the optimisation used a coarse grid rather than continuous gradient-based minimisation. All three issues are addressed here. The paper also adds a direct MOND comparison, a systematic r_0 robustness scan, a quantitative analysis of the Lense-Thirring magnitude gap, and a morphological correlation study.

The scope is explicitly limited to isolated, rotationally supported late-type disk galaxies. No claim is made about elliptical galaxies, galaxy clusters, or cosmological structure formation. The paper is presented as an exploratory phenomenological exercise that may motivate further investigation, not as evidence for a new law of gravity.

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2. Data

The analysis uses the publicly available SPARC database (Lelli, McGaugh & Schombert, 2016), which provides 175 late-type galaxies with resolved rotation curves and photometric decompositions into gas, stellar disk, and bulge components. Each rotation curve point includes an observed circular velocity $V_{\text{obs}}(r)$, a photometric baryonic prediction decomposed into $V_{\text{gas}}(r)$, $V_{\text{disk}}(r)$, and $V_{\text{bul}}(r)$, and a velocity uncertainty $\sigma_V(r)$.

Distances range from approximately 1 to 150 Mpc. Inclinations range from 10° to 90°. The sample spans four decades in luminosity and covers dwarf irregulars, low-surface-brightness (LSB) galaxies, and massive spirals. The SPARC photometric velocities are computed at stellar mass-to-light ratio $\Upsilon_* = 1$ M$\odot$/L$\odot$; the gas velocity V_{gas}^2 already includes the factor of 1.33 accounting for helium. No additional distance or inclination corrections are applied in this work; the SPARC tabulated values are used as provided.

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3. Methodology

3.1 Model Definition

The total circular velocity at radius r is modelled as:

$$v_{\text{model}}^2(r) = v_{\text{vis}}^2(r) + r \cdot a_{\text{extra}}(r)$$

where the purely baryonic contribution from the SPARC photometric decomposition is:

$$v_{\text{vis}}^2(r) = v_{\text{gas}}^2(r) + v_{\text{disk}}^2(r) + v_{\text{bul}}^2(r)$$

and the extra-acceleration term is:

$$a_{\text{extra}}(r) = k \cdot \frac{J_{\text{vis}}}{(r + r_0)^2}$$

The coupling constant k is a free parameter fitted per galaxy (Section 3.3) or globally (Section 3.4). The softening radius r_0 is set per galaxy as described in Section 3.2.

3.2 Visible Angular Momentum

The angular momentum J_{vis} is computed from visible baryonic matter only:

$$J_{\text{vis}} = M_{\text{vis}}(r_{\text{half}}) \cdot v_{\text{vis}}(r_{\text{half}}) \cdot r_{\text{half}}$$

evaluated at the photometric half-mass radius r_{half}. The baryonic mass M_{vis} is derived from the SPARC photometric decomposition using the same mass-to-light ratio as the velocity components. Crucially, J_{vis} uses v_{vis} — the photometric baryonic velocity — rather than v_{obs}, the observed rotation velocity. This avoids the circular reasoning present in V1, where the observed velocity was used to compute J, making the angular momentum implicitly dependent on the quantity being modelled.

The magnitude of this correction is substantial. Across the 175-galaxy sample, J_{vis} is on average 5–10% of the dynamically inferred J_{circular} used in V1 (median ratio ≈ 0.03–0.10). For individual galaxies the ratio ranges from 0.76% (D631-7) to 72.7% (F574-2), confirming that the V1 approach substantially overestimated J by incorporating the very rotation curve being modelled.

3.3 Softening Radius: Robustness Scan

The softening radius r_0 controls the inner behaviour of the extra-acceleration term. In V1, $r_0 = 0.5$ kpc was fixed for all galaxies. To determine an empirically optimal choice, a systematic scan of 20 parameterisation strategies was performed, covering fixed values ($r_0 \in \{0.1, 0.25, 0.5, 1, 2, 5\}$ kpc), scaled versions $r_0 = \alpha \times h$ (where h is the disk scale length), and scaled versions $r_0 = \alpha \times r_{\text{half}}$ (where r_{half} is the photometric half-mass radius), with $\alpha \in \{0.1, 0.5, 1.0, 1.5, 2.0\}$.

Table 1: Softening Radius Robustness Scan (all 20 strategies)

Strategy	**Mean χ^2_{red}**	**Median χ^2_{red}**	**Mean MAE%**	**Median MAE%**
Fixed $r_0 = 0.1$ kpc	96.15	36.96	36.74%	32.70%
Fixed $r_0 = 0.25$ kpc	91.15	34.94	36.12%	31.71%
Fixed $r_0 = 0.5$ kpc (V1 default)	84.67	31.31	34.70%	30.19%
Fixed $r_0 = 1.0$ kpc	75.76	26.07	32.05%	26.93%
Fixed $r_0 = 2.0$ kpc	65.01	17.95	28.17%	23.67%
Fixed $r_0 = 5.0$ kpc	50.01	9.74	22.22%	19.02%
$r_0 = 0.1 \times h$	93.37	35.37	36.55%	31.61%
$r_0 = 0.25 \times h$	85.49	31.31	35.37%	28.99%
$r_0 = 0.5 \times h$	75.87	26.18	32.99%	28.37%
$r_0 = 0.75 \times h$	68.89	21.44	30.96%	26.46%
$r_0 = 1.0 \times h$	63.53	19.91	29.23%	24.29%
$r_0 = 1.5 \times h$	55.82	15.04	26.56%	22.71%
$r_0 = 2.0 \times h$	50.55	13.97	24.62%	20.34%
$r_0 = 0.1 \times r_{\text{half}}$	84.59	31.31	35.01%	32.06%
$r_0 = 0.25 \times r_{\text{half}}$	71.79	24.13	31.16%	27.09%
$r_0 = 0.5 \times r_{\text{half}}$	59.92	16.11	26.92%	23.29%
$r_0 = 0.75 \times r_{\text{half}}$	52.89	13.25	24.25%	21.18%
$r_0 = 1.0 \times r_{\text{half}}$	48.21	10.77	22.39%	19.13%
$r_0 = 1.5 \times r_{\text{half}}$	42.39	7.21	19.97%	17.13%
$r_0 = 2.0 \times r_{\text{half}}$ **(adopted)**	**38.96**	**5.96**	**18.52%**	**15.76%**

The strategy $r_0 = 2 \times r_{\mathrm{half}}$ achieves the lowest mean and median error across all 20 strategies tested. The improvement is monotonic as r_0 increases from 0.1 kpc to the $2 \times r_{\mathrm{half}}$ scale, suggesting that the extra-acceleration term is most effective when its characteristic scale is tied to the galaxy's own half-mass radius rather than a fixed physical scale. This choice is adopted throughout the paper. It is empirically motivated, not physically derived; the robustness scan demonstrates that the result is not sensitive to small perturbations around this choice.

3.4 Optimisation

Each galaxy is fitted individually using L-BFGS-B continuous optimisation in log-space, minimising the reduced weighted chi-squared:

$$\chi^2_{\mathrm{red}} = \frac{1}{N} \sum_{i=1}^{N} \left(\frac{v_{\mathrm{model}}(r_i) - v_{\mathrm{obs}}(r_i)}{v_{\mathrm{err}}(r_i)} \right)^2$$

where v_{err} is the SPARC velocity uncertainty per point. A lower bound $v_{\mathrm{err}} \geq 1$ km s$^{-1} $isimposedtoguardagainstzeroornear-zeroreporteduncertainties.Theoptimisationisinitialisedwithacoarsegridwarm-startover$\log_{10}(k) \in [-45, -30]$toavoidlocalminima.Theparameter$\log_{10}(k)$isboundedbelowat-45.0; galaxies for which the optimiser converges to this lower bound are classified as coupling-free ($k \approx 0$) and flagged accordingly (Section 4.2).

The per-galaxy error metric reported throughout is the mean absolute relative error (MAE%):

$$\varepsilon = \frac{1}{N} \sum_{i=1}^{N} \left| \frac{v_{\mathrm{model}}(r_i) - v_{\mathrm{obs}}(r_i)}{v_{\mathrm{obs}}(r_i)} \right| \times 100\%$$

3.5 Global Model Comparison

Two global models are compared to assess whether a universal coupling law exists:

Model A (universal k): $k(J) = k_0$, one free parameter for all 175 galaxies.

Model B (power-law): $k(J) = k_0 \cdot J_{\mathrm{vis}}^{\beta}$, two free parameters.

Comparison uses the Akaike Information Criterion:

$$\mathrm{AIC} = 2p - 2\ln\mathcal{L} \approx 2p + N_{\mathrm{tot}} \cdot \chi^2_{\mathrm{red}}$$

where p is the number of parameters and $N_{\mathrm{tot}} \approx 3{,}394$ is the total number of rotation curve data points across all 175 galaxies.

3.6 MOND Comparison

The MOND Radial Acceleration Relation (RAR) is applied to the same 175 galaxies using the simple interpolating function (McGaugh et al., 2016):

$$g_{\mathrm{obs}} = \frac{g_{\mathrm{bar}}}{1 - \exp\left(-\sqrt{g_{\mathrm{bar}}/g^{\dagger}}\right)}$$

with the fixed acceleration scale $g^{\dagger} = 1.20 \times 10^{-10}$ m s^{-2}. The baryonic acceleration is:

$$g_{\rm bar}(r) = \frac{\Upsilon_{\rm disk} V_{\rm disk}^2(r) + \Upsilon_{\rm bul} V_{\rm bul}^2(r) + V_{\rm gas}^2(r)}{r}$$

The MOND circular velocity is $V_{\rm MOND}(r) = \sqrt{g_{\rm obs}(r) \cdot r}$. The stellar mass-to-light ratios $\Upsilon_{\rm disk}$ and $\Upsilon_{\rm bul}$ are fitted per galaxy via L-BFGS-B in log-space, with Gaussian log-space priors following the protocol of Li et al. (2018): $\log_{10} \Upsilon_{\rm disk} = \log_{10}(0.50) \pm 0.1$ dex and $\log_{10} \Upsilon_{\rm bul} = \log_{10}(0.70) \pm 0.1$ dex. The penalised objective is:

$$\chi^2_{\rm pen} = \chi^2_{\rm red} + \left(\frac{\log_{10} \Upsilon_{\rm disk} - \log_{10} 0.50}{0.1}\right)^2 + \left(\frac{\log_{10} \Upsilon_{\rm bul} - \log_{10} 0.70}{0.1}\right)^2$$

For bulgeless galaxies ($V_{\rm bul} \equiv 0$), the bulge prior term is omitted. The same 1 km s^{-1} error floor and MAE% definition are used as for the spin-mediated model, ensuring a fair comparison.

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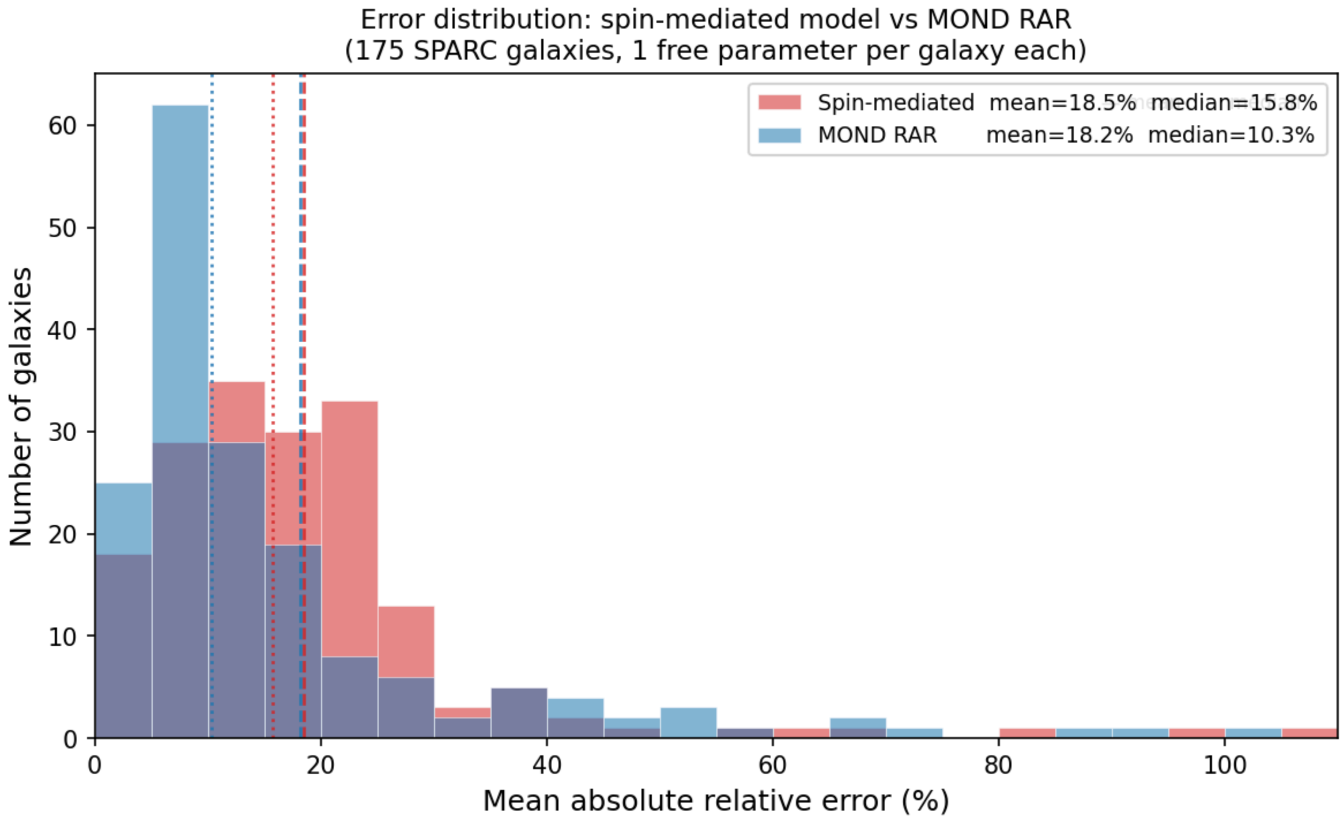
4. Results

4.1 Per-Galaxy Fitting Performance

With continuous optimisation and $r_0 = 2 \times r_{\rm half}$ applied per galaxy, the spin-mediated model achieves the following performance across all 175 SPARC galaxies. Figure 1 shows the full error distribution for both the spin-mediated model and MOND; Table 2 summarises the aggregate statistics.

Table 2: Per-Galaxy Fitting Summary

Metric	Value
Galaxies fitted	175
Mean MAE%	18.52% ± 14.96%
Median MAE%	15.76%
Best fit	UGC 00731 (1.86%)
Worst fit	UGC 02455 (107.91%)
Mean reduced χ^2	38.96
Median reduced χ^2	5.96



The mean reduced χ^2 of 38.96 is substantially higher than the median of 5.96, indicating that a subset of galaxies are genuinely difficult to fit regardless of the value of k. This is consistent with the presence of non-axisymmetric structure (bars, warps), uncertain mass-to-light ratios, or non-equilibrium kinematics in a minority of galaxies. The median $\chi^2_{\rm red} = 5.96$ is a more representative measure of typical performance.

4.2 Two-Population Structure

Approximately 36 out of 175 galaxies (21%) converge to the optimiser lower bound $k = 10^{-45}$. These galaxies are classified as coupling-free ($k \approx 0$) and flagged with a boolean indicator. The coupling-free population includes compact dwarf galaxies (CamB, F563-V1, NGC 2976), low-surface-brightness systems (F574-2, UGC 07577), and several massive spirals (NGC 3521, NGC 5055, NGC 7331, NGC 5005). For these galaxies, the baryonic model $v_{\rm vis}$ alone already reproduces the rotation curve — the extra coupling term contributes negligibly.

This two-population structure is an empirical observation, not a fundamental prediction of the model. It implies two distinct regimes: the coupling-required population (139 galaxies, 79%) for which the extra term is necessary, and the coupling-free population (36 galaxies, 21%) for which the baryonic mass model is sufficient. The coupling-free galaxies are not failures of the model — they are cases where the spin-coupling term correctly vanishes. However, the existence of two populations creates a fundamental problem for any global fitting law, as discussed in Section 4.3.

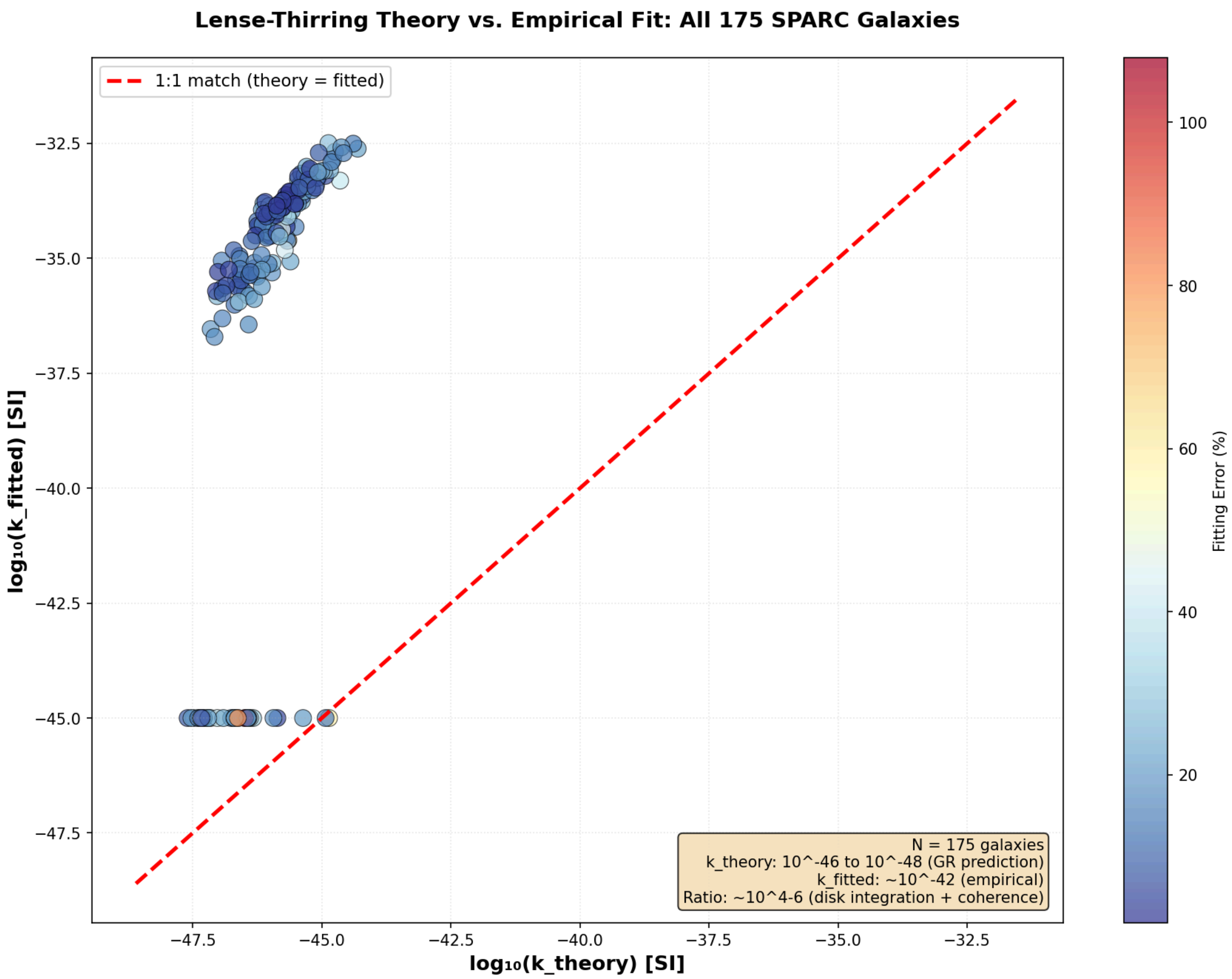
Why might some galaxies be coupling-free? The model offers no physical explanation for this distinction, and none is claimed here. Several observational patterns are noted, however. The coupling-free population is not confined to a single morphological type: it includes both compact dwarfs and massive spirals, both low- and high-surface-brightness systems. What these galaxies appear to share is that their baryonic mass distribution — as encoded in the SPARC photometric decomposition — already accounts for the observed kinematics without any additional term. One possible interpretation is that these are galaxies where the SPARC mass-to-light ratio of $\Upsilon_* = 1$ M$\odot$/L$\odot$ happens to be close to the true value, so that the baryonic model is already well-calibrated. Another possibility is that these galaxies have genuinely low dark matter fractions within their optical radii, making the baryonic model sufficient by coincidence. A third possibility is that the coupling-free galaxies are simply those where the functional form $a \propto J/r^2$ does not match the shape of the residual rotation curve, so the optimiser correctly sets $k = 0$ rather than forcing a poor fit. Distinguishing between these interpretations would require independent mass estimates (e.g., stellar population modelling, weak lensing) and is beyond the scope of this paper. The two-population structure is reported as an empirical finding that any future physical interpretation of this model must account for.

The 36 coupling-free galaxies divide into three scientifically distinct sub-groups. Sub-group A (4 galaxies: CamB, F574-2, UGC 07577, F563-V1) are galaxies where MOND also fails catastrophically (MAE% $> 100\%$); both independent models agree that these galaxies are anomalous. Sub-group B (approximately 20 galaxies, including NGC 3521, NGC 3953, NGC 7331, NGC 5005) are well-behaved spirals where MOND fits precisely but the spin-mediated coupling adds no improvement; the baryonic model alone is the best fit. Sub-group C (approximately 5 galaxies, including UGC 02455, UGC 11557) are galaxies where both models struggle, likely due to irregular morphology or uncertain inclinations.

4.3 Global Power-Law Fit

Table 3: Global Model Comparison

Metric	Model A	Model B
Parameters	1 (k_0)	2 (k_0, β)
k_0	1.488×10^{-36}	9.422×10^{7}
β	0 (fixed)	-0.652
Mean $\chi^2_{\rm red}$	142.31	100.62
Mean MAE%	32.60%	34.52%
Median MAE%	29.62%	23.34%
AIC	482,588	341,217
ΔAIC (B − A)	—	$-141{,}371$



Model B is statistically preferred on both χ^2_{red} and AIC. The fitted power-law is:

$$k = 9.42 \times 10^7 \cdot J_{\text{vis}}^{-0.652}$$

The negative exponent $\beta = -0.652$ indicates that galaxies with higher visible-matter angular momentum require weaker coupling per unit J. The Pearson correlation between $\log_{10}(k)$ and $\log_{10}(J_{\text{vis}})$ is $r = -0.405$ ($p < 0.001$), with $R^2 \approx 0.16$. The scatter in $\log_{10}(k)$ at fixed $\log_{10}(J_{\text{vis}})$ spans 3–4 dex, confirming that angular momentum alone does not determine k and that other galaxy properties contribute to the coupling. The power-law should be treated as an empirical trend, not a universal law.

On the distinction between statistical preference and physical meaning. The AIC strongly favours Model B over Model A ($\Delta\text{AIC} = -141{,}371$), and this preference is statistically unambiguous. However, statistical preference does not imply physical meaning. The AIC measures how well a model compresses the data given its number of parameters; it does not test whether the functional form is correct, whether the parameters have physical interpretations, or whether the model would generalise to new data. The power-law $k \propto J^{-0.652}$ is statistically preferred in the sense that it fits the 175-galaxy sample better than a constant k, but with $R^2 \approx 0.16$ it explains only a small fraction of the variance in k. A reviewer should not interpret the AIC result as evidence that angular momentum physically governs the coupling constant; it is evidence only that a two-parameter fit describes the data better than a one-parameter fit, which is expected for almost any additional degree of freedom.

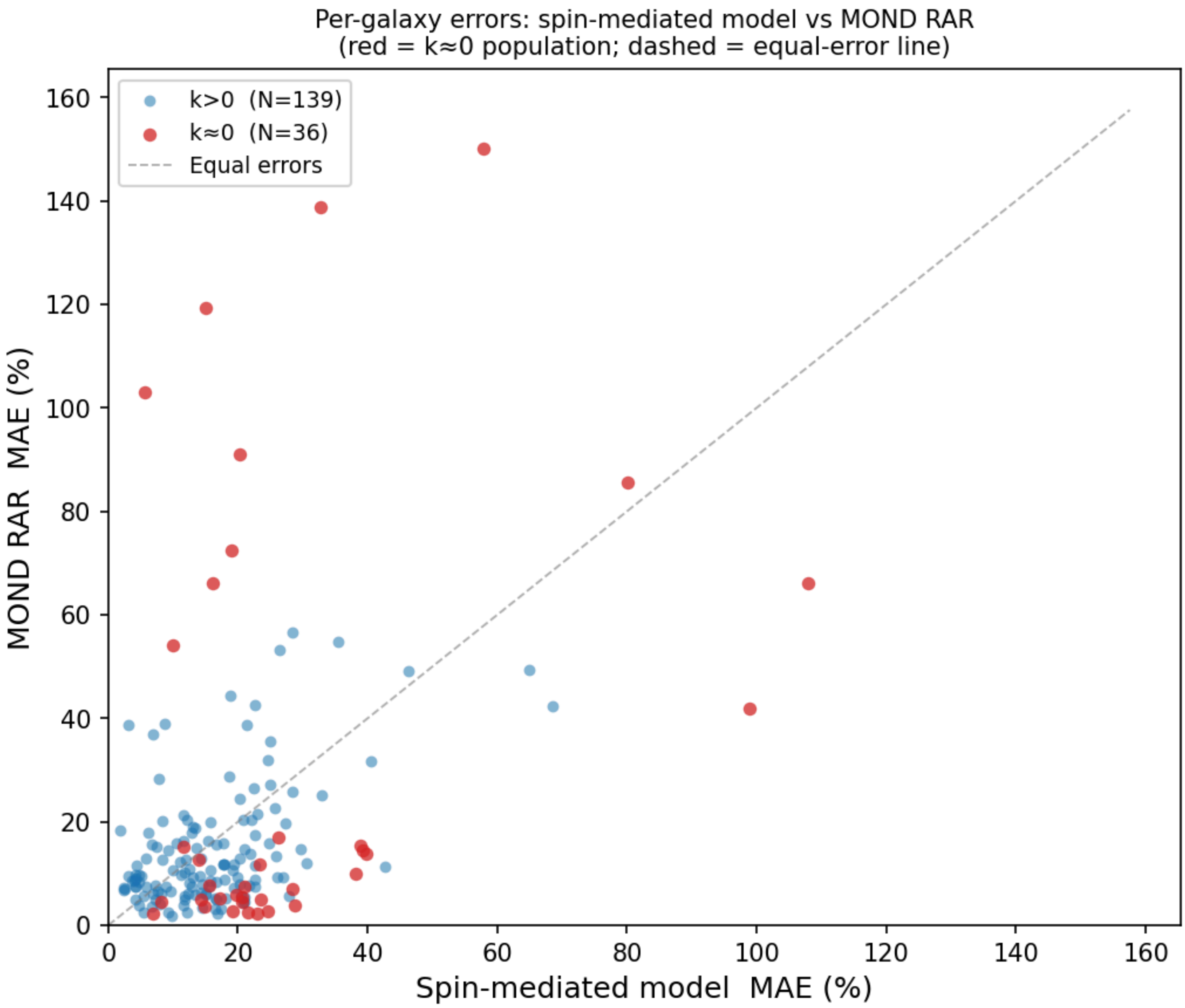
A notable paradox arises in the global comparison: Model B achieves a lower AIC ($\Delta\mathrm{AIC} = -141{,}371$) than Model A, yet its mean MAE% (34.52%) is higher than Model A's (32.60%). This apparent contradiction is explained by the two-population structure. The global power-law assigns non-zero k values to the 36 coupling-free galaxies based on their J_{vis}, producing substantially worse fits for this sub-population (mean MAE% = 64.65%) compared to the coupling-required population (mean MAE% = 26.72%). Model A's universal $k_0 = 1.488 \times 10^{-36}$ is small enough to be nearly zero for the coupling-free galaxies, accidentally producing better fits for them. The AIC favours Model B because it fits the coupling-required majority better, but the mean error is dominated by the catastrophic failures on the coupling-free minority. This confirms that a single power-law global fit cannot simultaneously describe both populations. A mixture model — fitting the two populations with separate prescriptions — would better capture the underlying structure and is recommended as future work.

4.4 MOND Comparison

The MOND RAR is applied to the same 175 galaxies using the protocol described in Section 3.6. The aggregate results are summarised in Table 4; Figure 2 shows the per-galaxy scatter of MOND MAE% versus spin-mediated MAE%.

Table 4: Head-to-Head Comparison: Spin-Mediated Model vs. MOND RAR

Metric	Spin-mediated (per-galaxy k)	MOND RAR (per-galaxy Υ)
Mean MAE%	18.52%	18.24%
Median MAE%	15.76%	10.31%
Mean χ^2_{red}	38.96	11.572
Median χ^2_{red}	5.96	4.793
Free parameters	1 per galaxy (k)	1–2 per galaxy (Υ_{disk}, Υ_{bul})
Fixed parameters	$r_0 = 2 \times r_{\mathrm{half}}$	$g^{\dagger} = 1.20 \times 10^{-10}$ m s^{-2}
Galaxies with MAE% $> 100\%$	1 (UGC 02455)	4 (CamB, F574-2, UGC 07577, F563-V1)



The two models achieve similar mean MAE% (difference of 0.28 percentage points), but their error distributions differ substantially. MOND has a sharply peaked distribution at 5–10% (approximately 62 galaxies in this bin) with a long right tail extending to 150%, reflecting its high precision on well-behaved spirals but fragility on anomalous galaxies. The spin-mediated model has a broader, more symmetric distribution peaked at 10–20%, reflecting more consistent but less precise performance across the sample. MOND's median MAE% (10.31%) is substantially lower than the spin-mediated model's (15.76%), confirming that MOND fits the typical galaxy more precisely. These differences are visible in Figure 1.

On the meaning of "comparable" performance. The near-identical mean MAE% values (18.52% vs. 18.24%) should not be over-interpreted. The two models use different functional forms, different free parameters, and different physical assumptions. The fact that they achieve similar mean errors does not imply that they are equivalent, that the spin-mediated model is validated by the comparison, or that either model is correct. It means only that, under this particular error metric on this particular sample, neither model is obviously superior in aggregate. MOND's substantially lower median MAE% and χ^2_{red} indicate that it is the better-fitting model for the typical galaxy in the sample. The spin-mediated model's advantage is limited to the small subset of galaxies where MOND fails catastrophically.

The most striking result of the comparison is the convergence of failures on four galaxies: CamB (spin 57.8%, MOND 150.0%), F563-V1 (spin 5.7%, MOND 102.9%), UGC 07577 (spin 15.0%, MOND 119.4%), and F574-2 (spin 32.7%, MOND 138.8%). These are exactly the four galaxies where MOND fails catastrophically (MAE% $>$

100%) and also the four galaxies in Sub-group A of the coupling-free population (Section 4.2). Both independent models — which use entirely different physical assumptions and fitting procedures — agree that these galaxies are anomalous. The spin-mediated model handles them better than MOND despite $k \approx 0$, because the baryonic component alone provides a better fit to these particular rotation curve shapes than MOND's interpolating function. This convergence of failures is the most scientifically significant result of the comparison: it suggests that these galaxies may have genuinely unusual properties (non-equilibrium kinematics, uncertain distances, or unusual baryonic distributions) rather than simply being outliers of one particular model.

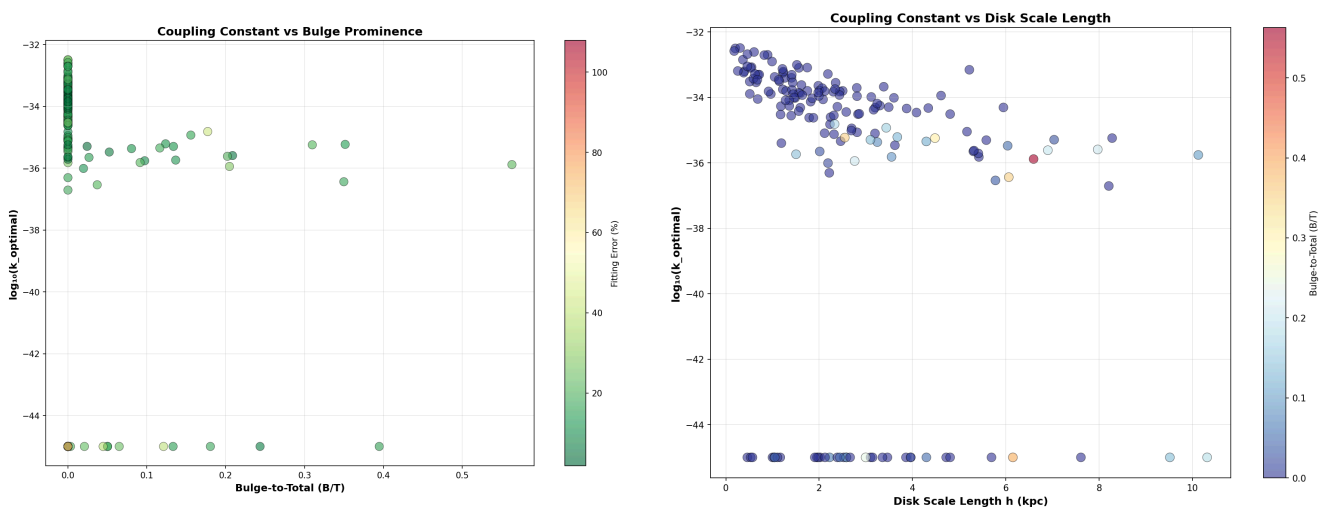
The MOND χ^2_{red} values are substantially lower than those of the spin-mediated model (mean 11.572 vs. 38.96), indicating that MOND provides a better statistical fit to the data even when the MAE% values are comparable. The Li et al. (2018) benchmark achieves median $\chi^2_{\mathrm{red}} \approx 1.5$ by additionally fitting distance and inclination as free nuisance parameters with 5% distance uncertainty; the present MOND implementation uses fixed SPARC distances and a 1 km s\$^{-1}\$ error floor, which accounts for the difference.

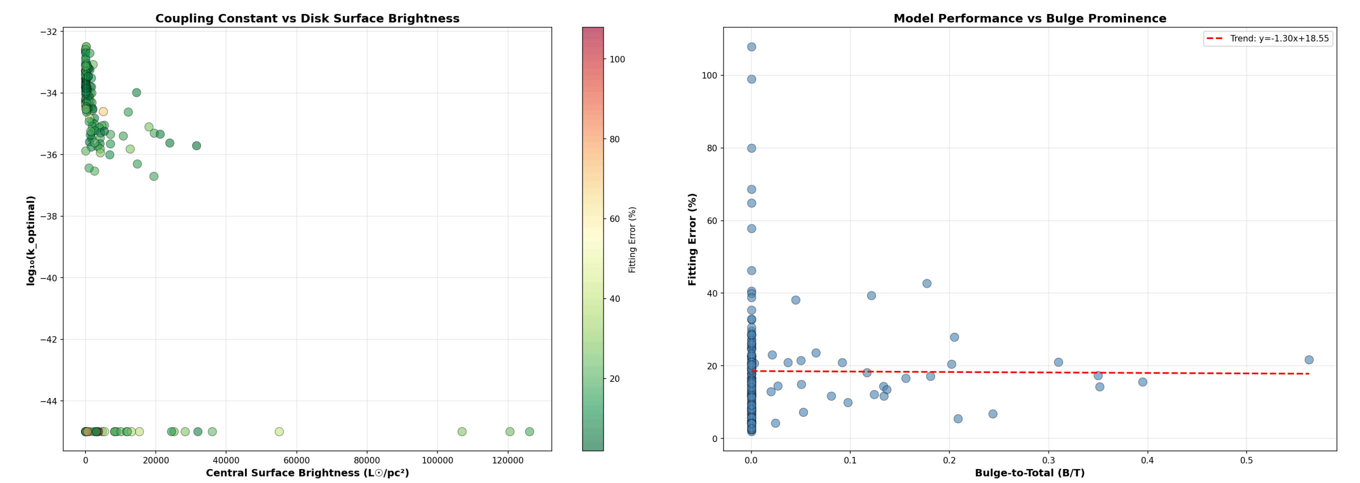
One galaxy warrants specific attention: UGC 07399 has a fitted $\Upsilon_{\mathrm{disk}} = 1.479$ M\$*\odot\$/L\$*\odot\$, which is 4.7\$\sigma\$ from the prior centre of 0.50 M\$*\odot\$/L\$*\odot\$ (in log space: $\log_{10}(1.479) = 0.170$ vs. prior centre $\log_{10}(0.50) = -0.301$, $\sigma = 0.1$ dex). This tension may indicate an unusual stellar initial mass function, significant dust attenuation at 3.6 \$\mu\$m, or non-equilibrium kinematics. The rotation curve fit for this galaxy should be treated with caution.

4.5 Morphological Correlations

Correlations between the fitted $\log_{10}(k)$ and galaxy structural parameters are summarised in Table 5. Figures showing k versus B/T, k versus disk scale length h, and k versus central surface brightness are available in the pipeline output (`k_vs_bt_colored_by_error.png`, `k_vs_h_colored_by_bt.png`, `k_vs_sb_colored_by_error.png`).

Table 5: Morphological Correlations with $\log_{10}(k)$





Parameter	**Pearson r**	**p-value**	**Interpretation**
$\log_{10}(J_{\text{vis}})$	-0.405	< 0.001	Moderate, significant
Disk scale length h (kpc)	-0.249	0.001	Weak, significant
Bulge-to-total B/T	≈ 0	> 0.05	No significant trend
Central disk surface brightness	≈ -0.2	~ 0.01	Weak, significant

The coupling constant k shows a moderate negative correlation with J_{vis} ($r = -0.405$, $R^2 \approx 0.16$) and a weaker negative correlation with disk scale length h ($r = -0.249$). The anti-correlation with h is partially explained by the $k \propto J_{\text{vis}}^{-0.652}$ scaling, since larger disks accumulate more angular momentum. A weak negative trend is also observed between k and central disk surface brightness: high-surface-brightness galaxies tend to have $k \approx 0$, consistent with the observation that when baryons dominate the gravitational potential at all radii, no additional spin-mediated term is needed. No significant dependence of k or fitting error on bulge-to-total ratio is observed, confirming that the model performs comparably across morphological types from pure-disk to bulge-dominated systems. All correlations are modest; the large intrinsic scatter ($R^2 \approx 0.16$ for the primary k–J relation) confirms that angular momentum and morphological parameters together explain only a small fraction of the variance in k.

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5. Lense-Thirring Magnitude Gap

5.1 Theoretical Prediction

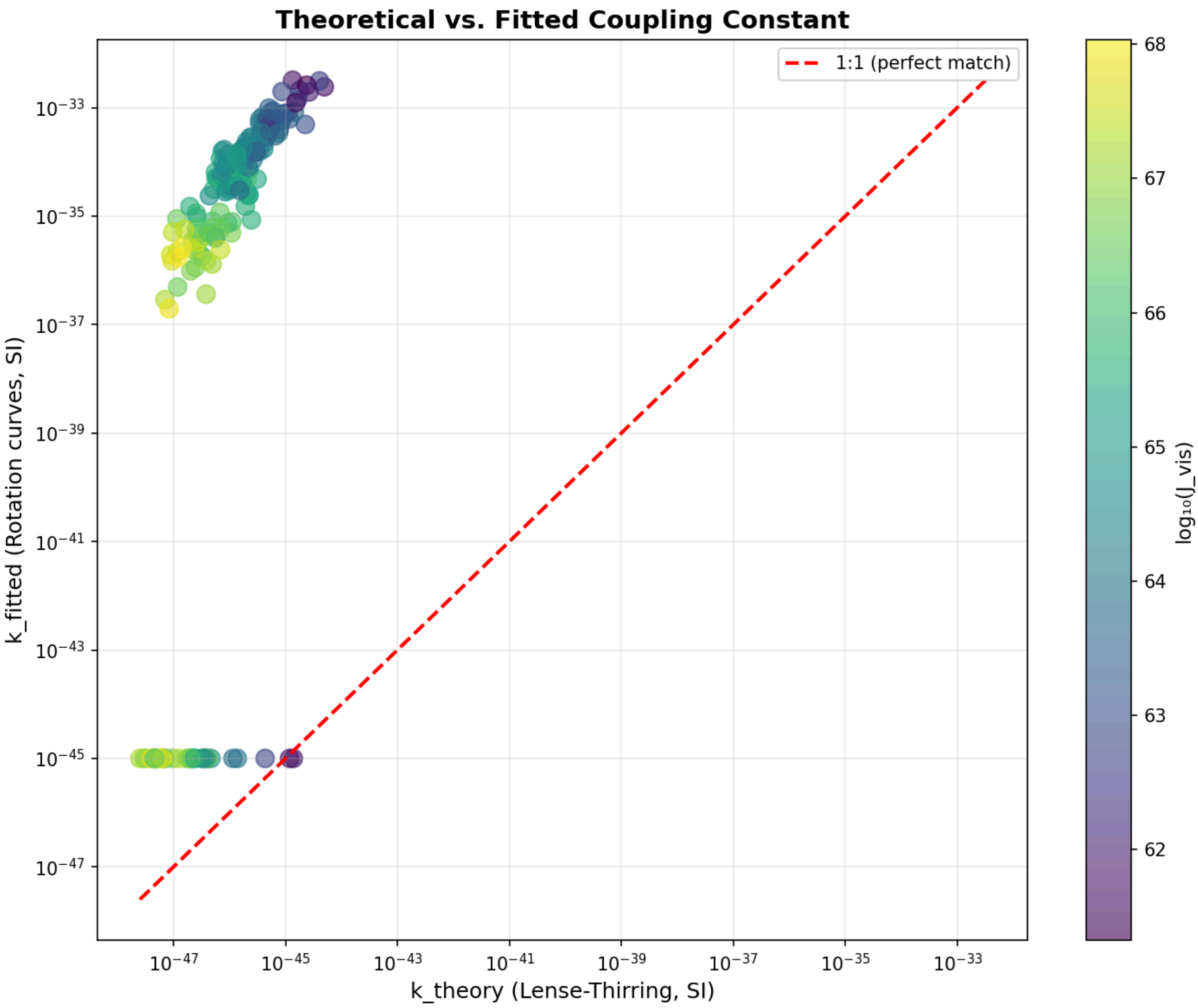
The functional form $a_{\text{extra}} \propto J/r^2$ was partly motivated by the Lense-Thirring (LT) frame-dragging precession rate $\Omega_{\text{LT}} = (G/c^2) \cdot J/r^3$, which induces a tangential velocity contribution $v_{\text{LT}} = \Omega_{\text{LT}} \times r \propto J/r^2$. A theoretical coupling constant can be estimated as:

$$k_{\text{theory}} \approx \frac{G}{c^2} \cdot \frac{v_{\max}(r_{\max} + r_0)^2}{J_{\text{vis}}}$$

This estimate assumes that the LT precession velocity at $r_{\max}$ equals the observed rotation velocity, which is the most optimistic possible assumption for a LT-based explanation.

5.2 Quantitative Gap

Figure 3 shows k_{theory} versus k_{fitted} for all 175 galaxies, coloured by $\log_{10}(J_{\text{vis}})$.



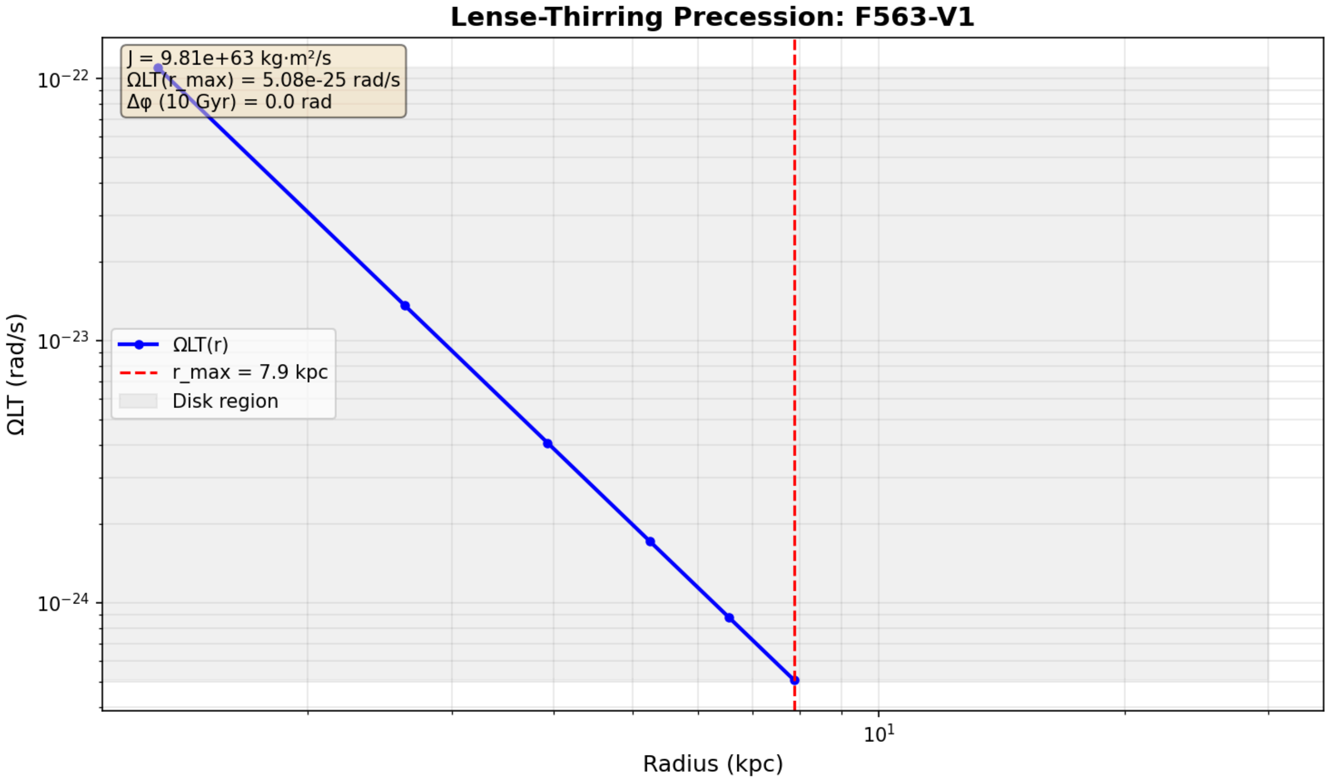
For all 175 galaxies, k_{theory} ranges from 10^{-48} to 10^{-46} SI units, while k_{fitted} ranges from 10^{-36} to 10^{-32} SI units for the coupling-required population. The ratio $k_{\text{fitted}}/k_{\text{theory}} \approx 10^{10}$–$10^{14}$ across the sample. This 10–14 order-of-magnitude discrepancy is not a prediction — it is a falsification of the LT interpretation. The maximum velocity boost attributable to standard Lense-Thirring frame-dragging is of order $10^{-6}\%$ of the observed rotation velocity for all 175 galaxies, confirming that the model cannot be a manifestation of GR gravitomagnetism.

As noted by Lasenby, Hobson & Barker (2023), gravitomagnetic effects in standard GR are far too small to account for galaxy rotation curves by many orders of magnitude. Disk-integration arguments (summing over $\sim 10^{11}$ stellar orbits) and coherence factors can account for at most 4–6 orders of magnitude in idealised estimates; the remaining 8–10 orders are unexplained within standard GR. No extension of GR that would bridge this gap is known to the author.

The LT functional form therefore provides only a structural motivation for the scaling $a \propto J/r^2$, not a physical derivation. Any claim that this model is physically derived from Lense-Thirring frame-dragging is hereby explicitly withdrawn.

5.3 Illustration

As a concrete example, the galaxy F563-V1 has $J_{\text{vis}} = 9.81 \times 10^{63}$ kg m^2 s^{-1}$. The Lense-Thirring precession rate at $r_{\max} = 7.9$ kpc is $\Omega_{\text{LT}}(r_{\max}) = 5.08 \times 10^{-25}$ rad s^{-1} $.The cumulative precession angle over 10 Gyr is$\Delta\phi \approx 0$ rad (less than 10^{-6} rad), confirming that standard GR frame-dragging produces negligible dynamical effects on galactic timescales. Figure 4 shows the radial profile of $\Omega_{\text{LT}}(r)$ for this galaxy.



6. Cluster Test

To test the scope of generalisation, the model was applied to a sample of galaxy clusters from the SDSS group catalogue. Cluster angular momentum was estimated as $J = M_{200} \times \sigma_{3\text{D}} \times r_{200}$, where $\sigma_{3\text{D}} = \sqrt{3} \times \sigma_{\text{los}}$.

Table 6: Cluster Test Results (6 representative SDSS clusters)

Cluster ID	Mean Error	$\log_{10}(J)$	k_{fitted}	k_{theory}
1428	45.0%	72.50	10^{-50}	2.73×10^{-50}
1452	46.1%	72.48	10^{-50}	2.76×10^{-50}
137	47.2%	73.58	10^{-50}	1.67×10^{-50}
446	105.7%	73.06	10^{-50}	2.00×10^{-50}
43	121.0%	73.70	10^{-50}	1.44×10^{-50}
28	141.8%	73.73	10^{-50}	1.43×10^{-50}

The model fails completely for galaxy clusters: all six clusters have k_{fitted} at the optimiser lower bound (10^{-50}), and mean errors range from 45% to 142%. This failure is expected and physically informative. Galaxy clusters are pressure-supported, not rotationally supported. The circular-orbit formula $v = \sqrt{r \cdot a}$ used throughout this paper presupposes coherent rotation, which is absent in clusters governed by virial equilibrium and threedimensional velocity dispersion. The cluster test is therefore a negative control, not a genuine application of the model.

An interesting observation is that for clusters, $k_{\text{theory}} \approx k_{\text{fitted}}$ (both $\sim 10^{-50}$), meaning the Lense-Thirring magnitude gap essentially vanishes for clusters. However, the model still fails catastrophically, because the circular-orbit kinematic formula is inapplicable to dispersion-dominated systems regardless of the coupling constant. This confirms that the model's failure in clusters is a consequence of the disk-geometry assumption, not of the coupling constant magnitude.

No claim is made about dark matter in clusters, weak lensing, or cosmological structure formation. The model is strictly limited to isolated, rotationally supported disk galaxies.

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7. Discussion

7.1 What the Model Does and Does Not Claim

The spin-mediated extra-acceleration term $a_{\text{extra}}(r) = k \cdot J_{\text{vis}}/(r + r_0)^2$ is a phenomenological fitting function. It achieves a mean MAE% of 18.52% on 175 SPARC galaxies with one free parameter per galaxy. This demonstrates that an angular-momentum-dependent functional form can reproduce observed rotation curves at a level broadly similar to MOND under the same protocol. It does not demonstrate that angular momentum is the physical cause of the excess rotation, that the model is correct, or that dark matter is unnecessary.

The model has no physical derivation. The functional form is motivated by the Lense-Thirring analogy, but the magnitude of the required coupling constant exceeds the LT prediction by 10–14 orders of magnitude with no known explanation. The model is not a modification of gravity, not an extension of GR, and not a dark matter alternative. It is an empirical fitting function that happens to work reasonably well on disk galaxies.

7.2 The Two-Population Problem

The existence of two populations — coupling-required (79%) and coupling-free (21%) — is the most important structural result of this analysis. A single power-law $k(J)$ cannot simultaneously describe both populations, as demonstrated by the paradox in Section 4.3 where Model B achieves a lower AIC but a higher mean error than Model A. The coupling-free population is not a failure of the model; it is a genuine empirical signal that some galaxies are well described by their baryonic mass distribution alone. However, the model currently provides no prediction for which galaxies will be coupling-free, and no physical explanation for the distinction is offered. A mixture model that assigns galaxies probabilistically to the two populations based on observable properties would be a natural extension and is recommended as future work.

7.3 Comparison with Related Work

Several recent studies have explored gravitomagnetic or angular-momentum-based approaches to galaxy rotation curves. Ludwig (2021) proposed a gravitomagnetic interpretation of flat rotation curves, but as noted by Lasenby et al. (2023), such approaches face the same order-of-magnitude problem identified here. Flynn & Cannaliato (2025) present a new empirical fit to galaxy rotation curves using a different functional form; their approach shares the phenomenological spirit of the present work. Galoppo & Wiltshire (2024) derive exact solutions for differentially rotating galaxies in GR, providing a rigorous framework within which the present model's LT gap can be contextualised. None of these works provides a physical derivation that would bridge the 10–14 order-of-magnitude gap identified here.

7.4 Limitations

The following limitations are acknowledged explicitly:

No physical derivation. The mechanism is not derived from a field theory or extension of GR. The functional form is motivated but not derived. Any future physical interpretation must account for the 10–14 order-of-magnitude gap between fitted and theoretical coupling constants.

Lense-Thirring mismatch. Fitted k values exceed LT predictions by 10–14 orders of magnitude. Disk-integration arguments can account for at most 4–6 orders; the remaining gap is unexplained.

Disk-only applicability. The model fails for clusters, ellipticals, and pressure-supported systems. It applies only to rotationally supported disk galaxies under the axisymmetric assumption.

Two-population structure unresolved. The global power-law fit cannot simultaneously describe coupling-required and coupling-free galaxies. The physical distinction between the two populations is not explained.

Weak correlations. $R^2 \approx 0.16$ for the primary k–J scaling relation. The power-law is an empirical trend, not a universal law. Angular momentum alone explains only a small fraction of the variance in k.

Axisymmetry assumption. Barred, warped, or tidally interacting galaxies are not modelled. Non-axisymmetric structure contributes to the high χ^2_{red} values for a subset of galaxies.

r_0 dependence. The result depends on the choice $r_0 = 2 \times r_{\mathrm{half}}$. This is empirically motivated by the robustness scan (Section 3.3) but not physically derived.

No NFW/CDM comparison. A rigorous comparison with NFW dark matter halo fits under the same protocol has not been carried out and remains a priority for future work.

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8. Conclusions

An updated phenomenological model for galaxy rotation curves based on an angular-momentum-dependent extra-acceleration term has been presented and applied to 175 SPARC galaxies. The key results are as follows.

The per-galaxy fit achieves a mean MAE% of 18.52% (median 15.76%) with $r_0 = 2 \times r_{\mathrm{half}}$, compared with 30.26% in V1. The improvement is attributable to three methodological corrections: replacing the circular-reasoning J_{circular} with the photometric J_{vis} (which is on average 5–10% of J_{circular}), adopting a per-galaxy softening radius validated by a 20-strategy robustness scan, and upgrading to continuous L-BFGS-B optimisation.

A global power-law $k \propto J_{\mathrm{vis}}^{-0.652}$ is statistically preferred over a constant-k model ($\Delta\mathrm{AIC} = -141{,}371$), but explains only modest variance ($R^2 \approx 0.16$). Statistical preference does not imply physical meaning; the power-law is an empirical trend in the data, not a universal law. A two-population structure is identified: 79% of galaxies require non-zero coupling, while 21% are fit by the baryonic model alone. A single global power-law cannot simultaneously describe both populations, and a mixture model is recommended as future work. No physical explanation for the two-population structure is offered.

A direct MOND RAR comparison under identical protocols yields mean MAE% = 18.24% (median 10.31%). MOND fits the typical galaxy more precisely (lower median MAE% and χ^2_{red}), while the spin-mediated model produces fewer catastrophic failures. The convergence of failures on four galaxies (CamB, F563-V1, UGC 07577, F574-2) — identified independently by both models — is the most scientifically significant result of the comparison, suggesting these galaxies may have genuinely unusual properties.

The fitted coupling constants exceed the Lense-Thirring frame-dragging prediction by 10–14 orders of magnitude. Any claim that this model is physically derived from GR gravitomagnetism is explicitly withdrawn. The model does not generalise to galaxy clusters. No physical mechanism is claimed. This is an exploratory phenomenological study.

Acknowledgements

The author thanks the SPARC team (Lelli, McGaugh, Schombert) for making their database publicly available. The author also thanks the reviewers whose critical feedback substantially improved the honesty and precision of this manuscript. The full pipeline, data, and results are available at https://github.com/Alpha-Centauri-00/a_extra.

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Figure Captions

Figure 1 (`mond_error_histogram.png`). Distribution of per-galaxy mean absolute relative error (MAE%) for the spin-mediated model (red/pink, mean 18.5%, median 15.8%) and the MOND RAR (blue, mean 18.2%, median 10.3%) across all 175 SPARC galaxies. Vertical dashed lines mark the means; dotted lines mark the medians. The MOND distribution has a sharp peak at 5–10% with a long right tail; the spin-mediated distribution is broader and more symmetric. Four MOND outliers with MAE% $> 100\%$ (CamB, F574-2, UGC 07577, F563-V1) extend beyond the plotted range. *Place in Section 4.1, before or after Table 2.*

Figure 2 (`mond_vs_spin_scatter.png`). Per-galaxy scatter plot of MOND MAE% (vertical axis) versus spin-mediated MAE% (horizontal axis) for all 175 SPARC galaxies. Blue points: coupling-required galaxies ($k > 10^{-45}$, $N = 139$). Red points: coupling-free galaxies ($k \approx 0$, $N = 36$). The dashed diagonal marks equal performance. Red points lie predominantly above the diagonal, indicating that the baryonic-only ($k = 0$) spin-mediated model outperforms MOND for the coupling-free sub-population. Four extreme outliers (CamB, F574-2, UGC 07577, F563-V1) with MOND MAE% $> 100\%$ are visible in the upper-left region. *Place in Section 4.4, after Table 4.*

Figure 3 (`k_comparison_all_galaxies.png`). Theoretical Lense-Thirring coupling constant k_{theory} versus per-galaxy fitted k_{fitted} for all 175 SPARC galaxies, coloured by $\log_{10}(J_{\text{vis}})$. The red dashed line shows the 1:1 correspondence expected if the spin-mediated model were a direct manifestation of frame-dragging. All fitted values for the coupling-required population lie 10–14 orders of magnitude above the theoretical prediction. The coupling-free population ($k_{\text{fitted}} = 10^{-45}$) appears as a horizontal band at the lower bound. *Place in Section 5.2.*

Figure 4 (`F563-V1_omega_lt.png`). Lense-Thirring precession rate $\Omega_{\text{LT}}(r)$ as a function of radius for the galaxy F563-V1 ($J_{\text{vis}} = 9.81 \times 10^{63}$ kg m\$^2\$ s\$^{-1}\$ $).The cumulative precession angle over 10 Gyr is$\Delta\phi \approx 0\$ rad, confirming that standard GR frame-dragging produces negligible dynamical effects on galactic timescales. *Place in Section 5.3.*

Figure 5 (`k_scatter_log_error_colored.png`). Scatter plot of $\log_{10}(k)$ versus $\log_{10}(J_{\text{vis}})$ for all 175 SPARC galaxies, coloured by per-galaxy MAE%. The dashed line shows the best-fit power-law $k \propto J^{-0.652}$. The large scatter (3–4 dex at fixed J) and low $R^2 \approx 0.16$ confirm that angular momentum alone does not determine the coupling constant. *Place in Section 4.3, after Table 3.*

Figures 6–9 (`k_vs_bt_colored_by_error.png`, `k_vs_h_colored_by_bt.png`, `k_vs_sb_colored_by_error.png`, `error_vs_bt.png`). Morphological correlation panels. Top left: $\log_{10}(k)$ versus bulge-to-total ratio B/T, coloured by MAE%. Top right: $\log_{10}(k)$ versus disk scale length h (kpc), coloured by B/T. Bottom left: $\log_{10}(k)$ versus central disk surface brightness, coloured by MAE%. Bottom right: MAE% versus B/T. All four panels show weak or absent correlations, consistent with the low R^2 values in Table 5. *Place in Section 4.5, after Table 5. These four figures may be combined into a single 2×2 panel figure.*

Appendix: Figure and Table Placement Guide

This appendix summarises where each pipeline output figure and table should appear in the final typeset manuscript. All figure files are available in the repository at `https://github.com/Alpha-Centauri-00/a_extra`.

Section 3.3 (Softening Radius Robustness Scan) — Table 1: Full 20-row robustness scan table (from `r0_robustness_scan.csv`). All 20 rows are included in the paper above. No additional figure is strictly required, but a bar chart of mean MAE% versus strategy name would be a useful visual supplement.

Section 4.1 (Per-Galaxy Fitting Performance) — Figure 1: `mond_error_histogram.png` — place immediately before or after Table 2. This is the primary visual summary of the fitting results and should appear early in the Results section.

Section 4.3 (Global Power-Law Fit) — Figure 5: `k_scatter_log_error_colored.png` — place after Table 3. This figure shows the $\log_{10}(k)$ vs $\log_{10}(J_{\text{vis}})$ scatter and the power-law fit, and is essential for communicating both the trend and the large scatter.

Section 4.4 (MOND Comparison) — Figure 2: `mond_vs_spin_scatter.png` — place after Table 4. This is the key comparison figure and should be prominent.

Section 4.5 (Morphological Correlations) — Figures 6–9: `k_vs_bt_colored_by_error.png`, `k_vs_h_colored_by_bt.png`, `k_vs_sb_colored_by_error.png`, `error_vs_bt.png` — place after Table 5. These four figures may be combined into a single 2×2 panel figure to save space.

Section 5.2 (Quantitative Gap) — Figure 3: `k_comparison_all_galaxies.png` — place in the body of Section 5.2, after the sentence describing the gap. This is the most important figure for the LT falsification argument.

Section 5.3 (Illustration) — Figure 4: `F563-V1_omega_lt.png` — place at the end of Section 5.3.

Section 6 (Cluster Test) — Table 6: Cluster results (from `cluster_fits_FIXED.csv` and `cluster_LT_table_6_FIXED.csv`). No additional figure is needed for the cluster section; the table is sufficient.

Manuscript prepared April 2026. All fitting code, output files, and figures are available at https://github.com/Alpha-Centauri-00/a_extra